

2nd Wind

FOH CHANNEL EQ SETUP — DiGiCo Quantum 225

Show	2nd Wind
Show Date	2026-08-08
Venue	Fountain Square (outdoor)
Console	DiGiCo Quantum 225
Channels	24
FOH Engineer	Brian Lloyd
Monitor Engineer	—
Show Time	7:00pm–10:00pm
Rev	Rev 2.0 Deep Build

DOCUMENT CONTENTS

1. Cover Page — show, venue, and personnel
2. Input List — color-coded full channel patch sheet
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SHOW STYLE: Fifth & Vine Live, 7–10 pm on the plaza. Four handheld vocals over a rehearsed R&B/funk show band running to a click, in 93 F air at 38 percent humidity. Clarity first, cuts before boosts, and every dB of vocal presence bought by clearing the instruments rather than by lifting the voice.

2nd Wind

VENUE: Fountain Square (outdoor)

DATE: 2026-08-08

REV: Rev 2.0 Deep Build

FOH: Brian Lloyd

MON: —

SHOW TIME: 7:00pm–10:00pm

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
■ DRUMS						
1	Kick In	Shure Beta 91A	Local 1	✓	—	Boundary mic inside the shell. Contour switch ASSUMED FLAT — if the low-mid scoop is engaged the capsule is already doing –7dB at 400, so halve B3 to –4.
2	Kick Out	Shure Beta 52A	Local 2		Short	At the sound hole / just outside the resonant head. Mic changed from rev 1's D6.
3	Snare Top	Audix i5	Local 3		Short	Locker fork — Beta 98H/C was offered and the i5 taken on 2026-07-31; Brian's call stands, not re-litigated.
4	Snare Bottom	Electro-Voice N/D 408	Local 4		Clip	■ POLARITY INVERT against ch 3 — not optional. Claw / rim mount. Dynamic: 48V stays OFF. A bare '408' is always the EV N/D 408 (Brian, 2026-08-08).
5	Hat	Shure SM81	Local 5	✓	Boom	One of three SM81s on this kit (5 / 14 / 19). Onboard low-cut ASSUMED 18dB/oct engaged, pad OUT — if left flat the desk HPF carries alone and there is more rumble to ride. Mic changed from rev 1's ND408.
6	Rack 1	Audix D2	Local 6		Clip	Highest of three toms, each on a different capsule (D2 / D4 / D6).
7	Rack 2	Audix D4	Local 7		Clip	Mic changed from rev 1's second D2 — re-derived from scratch, the D4's curve is nothing like the D2's.
8	Floor	Audix D6	Local 8		Clip	Locker fork — MD 421-U offered, D6 KEPT (Brian, 2026-08-08). Built for the scooped voicing. Mic changed from rev 1's D4.

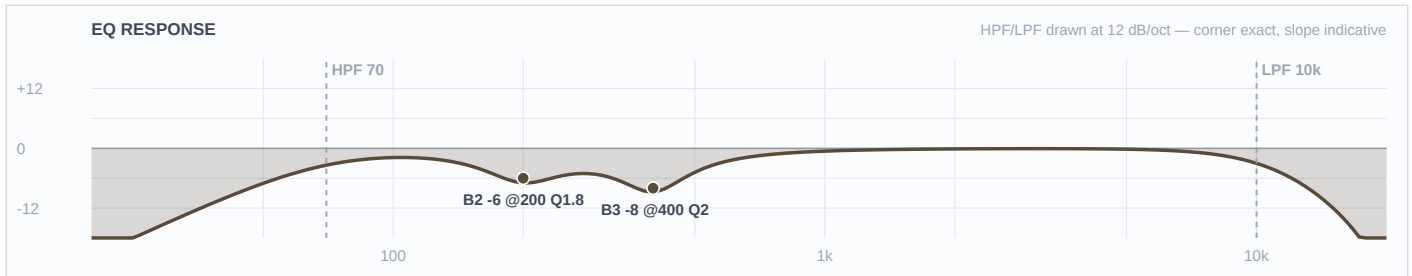
Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
15	Click	XLR line feed	Local 15		—	■ MUTED FROM THE MAINS. OUT OF THE RECORD BUS. Band-provided feed. Confirmed a real input (Brian, 2026-08-08) — they run to a click.
16	Track	XLR line feed	Local 16		—	Band-provided feed — confirm at load-in. MONO (confirmed).
17	Pad 1	BSS AR-133 (DI)	Local 17	✓	DI	Sampling pad run as TWO MONO channels panned, not a stereo image (Brian, 2026-07-31 — carried, not re-asked). Renumbered from rev 1's fader 21.
18	Pad 2	Whirlwind IMP (passive DI)	Local 18		DI	Second output of the same pad, panned opposite ch 17. Renumbered from rev 1's fader 22.
■ DRUMS						
19	Toys	Shure SM81	Local 19	✓	Boom	The only hand percussion on this show — rev 1's congas x2 and bongos are gone. Third of three SM81s (5 / 14 / 19). Onboard low-cut ASSUMED 18dB/oct engaged, pad OUT. Renumbered from rev 1's fader 20.
■ KEYS						
20	Key L	XLR line feed	Local 20		—	Band-provided feed — confirm at load-in. Stereo pair with ch 21. Renumbered from rev 1's fader 23.
21	Key R	XLR line feed	Local 21		—	Band-provided feed — confirm at load-in. Stereo pair with ch 20. Renumbered from rev 1's fader 24.
■ VOCALS						
33	Aretha	Shure Beta 58A (Wireless 1)	Local 33		—	Wireless 1. Windscreen — 15–16mph gusts all evening. RING OUT AT SOUNDCHECK: this MD removes the template's HPF 184, its –18@5k Q20 notch and its –6.3@335.

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
34	Heather	Shure Beta 58A (Wireless 2)	Local 34		—	Wireless 2. Windscreen. RING OUT AT SOUNDCHECK — template baseline removed.
35	Vince	Shure Beta 58A (Wireless 3)	Local 35		—	Wireless 3. Windscreen. RING OUT AT SOUNDCHECK. Fundamentals reach ~82Hz — the template's HPF 184 is flatly wrong for this voice and is removed.
36	Brandon	Shure Beta 58A (Wireless 4)	Local 36		—	■ LOAD-IN FLAG — NEW vocalist, range unknown (replaces Markay). Wireless 4. Windscreen. RING OUT AT SOUNDCHECK. ALTERNATES, printed here for a 20-second change once you hear him: if BARITONE (like Vince) HPF 95 · B3 -4@750 · B2 -7@400. If TENOR (like Markay) HPF 120 · B3 -3@1400 · B2 -6@500.

■ DRUMS ■

Ch 1 | Kick In

Mic: Shure Beta 91A



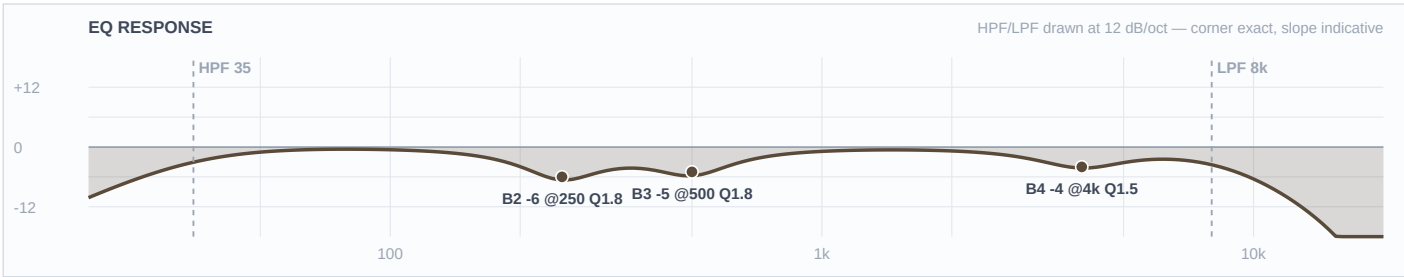
Mic Notes: Shure/Gear4Music: emphasis from 4kHz to ~9kHz with a presence boost around 7kHz — the beater attack is BAKED IN, which is why B4 is flat rather than boosted. Two-mic pair with ch 2: CH 1 OWNS ATTACK/TOP, ch 2 owns body/thump/bottom. Low end handed off deliberately — HPF 70 and B1 flat. Polarity-check the pair in mono at soundcheck.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	70 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	400 Hz	Bell	-8 dB	2	
2	200 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: This is the attack half of a two-mic kick on a plaza dance floor, and its whole job is to be heard at the back of the square while ch 2 supplies what the crowd feels up front. The 4–9kHz beater snap is Shure's own voicing, so rev 1's +3@3500 came out — a Q1.5 bell there reaches into that emphasis and stacks on it, and at 38 percent humidity the air is already taking some of that top, so trimming it would be wrong in the other direction. What's left is the two problems the capsule doesn't solve: -8@400 for the shell box a boundary mic collects, at outdoor depth because a plaza gives nothing back, and -6@200 for the cardboard that would otherwise pile onto the 52A's low reach.

Ch 2 | Kick Out

Mic: Shure Beta 52A



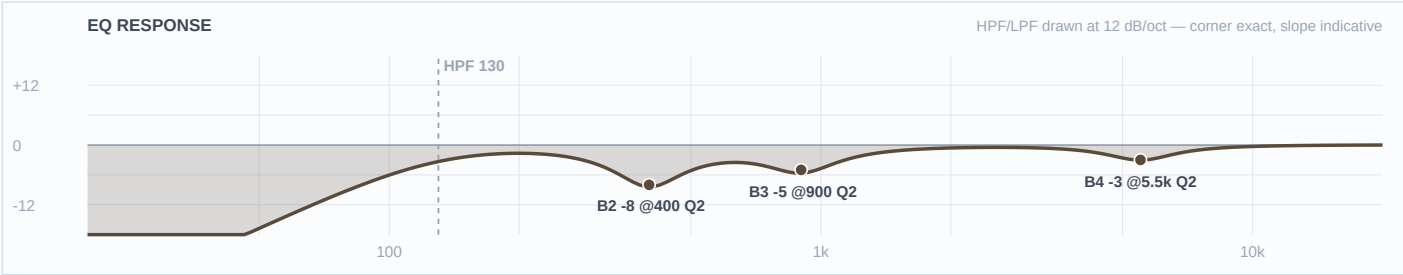
Mic Notes: Shure: presence boost at 4kHz, response deliberately limited to 20Hz–10kHz, emphasis across 50Hz–4kHz — so the capsule is already a low-pass filter and its weight is voiced in. Shure's own how-to names this exact pairing (91A inside, 52A at the sound hole). Two-mic pair with ch 1: CH 2 OWNS BODY/THUMP/BOTTOM. B1 flat because a low boost would stack on voiced response; B4 trimmed to keep out of ch 1's attack lane. Polarity-check in mono; flip this channel if the sum thins.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	35 Hz	LC	—	—	High-pass
4	4 kHz	Bell	-4 dB	1.5	
3	500 Hz	Bell	-5 dB	1.8	
2	250 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

Engineer Notes: The weight half of the kick, and on a plaza with 8 KS21s in a delayed arch this is the channel the dance floor actually feels. Everything here is a refusal or a de-stack. The forums' standard +2@60 is refused outright — Shure emphasises 50Hz–4kHz on purpose, so the bottom is claimed by gain and placement instead. The -4@4000 is not a tone decision but lane discipline: this capsule bakes a presence peak at 4k and ch 1 bakes an emphasis from 4–9k, so two voiced peaks on one drum in one zone is exactly the two-mic failure, and ch 1 wins that lane. B3 is only -5 despite the outdoor override because the capsule already scoops low-mids and deep-cutting a scoop digs a hole twice. LPF 8000 earns its place rejecting bleed from three cymbal mics and a bottom-snare.

Ch 3 | Snare Top

Mic: Audix i5



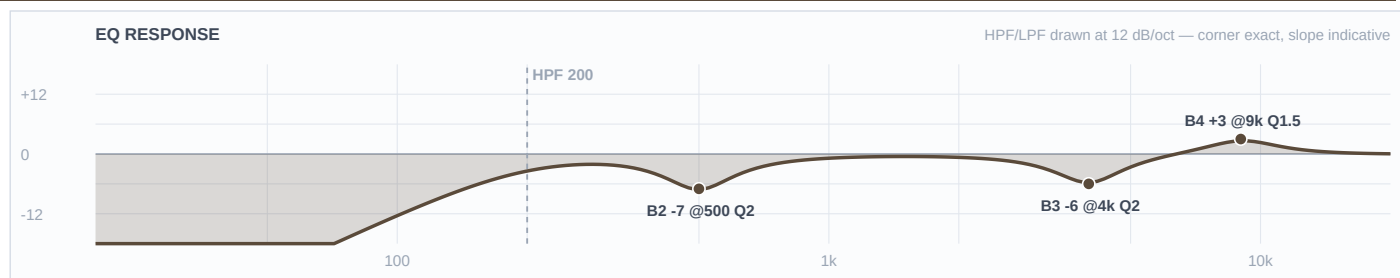
Mic Notes: RecordingHacks / SOS / Barry Rudolph: +5dB at 150Hz and +9dB at 5500Hz, with a broader presence lift across 3–8kHz. BOTH reflexes are therefore reversed — no crack boost, no body boost. HPF is set at 130 rather than 150 on purpose, so the filter doesn't claw back the baked body lift this channel owns. Two-mic pair with ch 4: CH 3 OWNS CRACK (5.5k, baked) AND BODY (150, baked); ch 4 owns the wires. Ch 4 is polarity inverted against this channel.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	130 Hz	LC	—	—	High-pass
4	5.5 kHz	Bell	-3 dB	2	
3	900 Hz	Bell	-5 dB	2	
2	400 Hz	Bell	-8 dB	2	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: The two-and-four the crowd moves to, so this is the single most consequential channel on the kit. The genre says boost the crack and the mic already did it by +9dB, so the desk's move there is a trim — and at 38 percent humidity, with dry air eating high frequencies all the way to the back of the plaza, that trim is -3 rather than the -4 it was last week at 50 percent and rising. Same drum, same capsule, different air, and that is the only reason the number moved. The -5@900 is deliberately lighter than outdoor default because the i5's mids sit slightly scooped against a 57 already, while -8@400 gets the full outdoor treatment because shell box is a problem no capsule helps with and a plaza has no room gain to hide it. Ghost notes and press rolls have to survive, which is another reason nothing here is aggressive above 5k.

Ch 4 | Snare Bottom

Mic: Electro-Voice N/D 408



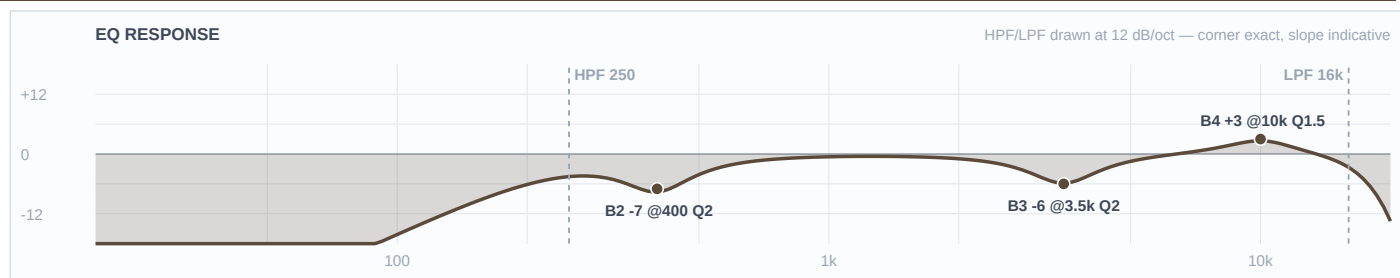
Mic Notes: ■ POLARITY INVERT. SOS and Sweetwater: a stick hit moves the top head AWAY from the top mic while driving the bottom head TOWARD this one, so it inverts relative to ch 3 every time — and what it hears is the snare wires, which 'add brilliance and bite to the snare sound picked up from above.' EV N/D 408: supercardioid N/DYM, 30Hz–22kHz, brighter and more aggressive UPPER MIDS than a 57 with a deliberate proximity lift. That aggression is trimmed at 4000 and the +3 sits at 9000, where neither EV's spec nor the KB documents a peak. Two-mic pair with ch 3: CH 4 OWNS WIRE SIZZLE ONLY, no body — HPF 200 and B1 flat.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	200 Hz	LC	—	—	High-pass
4	9 kHz	Bell	+3 dB	1.5	
3	4 kHz	Bell	-6 dB	2	
2	500 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: One job: the wires. In funk and Motown the articulation is the music — ghost notes, press rolls, rimshots — and this is the channel where those read, layered under ch 3's crack rather than competing with it. The +3@9000 is the only real boost on the whole kit and it's earned twice over: the capsule has no documented peak up there, and it is the one zone ch 3 doesn't touch, so it stacks with nothing. The -6@4000 does double duty, taming the 408's spiky upper mids and keeping this channel out of ch 3's baked 5.5k crack lane. The claw mount puts it very close to the head, which is exactly what piles low-mids, so HPF runs high at 200 and the box cut sits at 500 — offset from ch 3's 400 on purpose, because two mics on one drum digging the identical hole is how a snare ends up hollow.

Ch 5 | Hat

Mic: Shure SM81



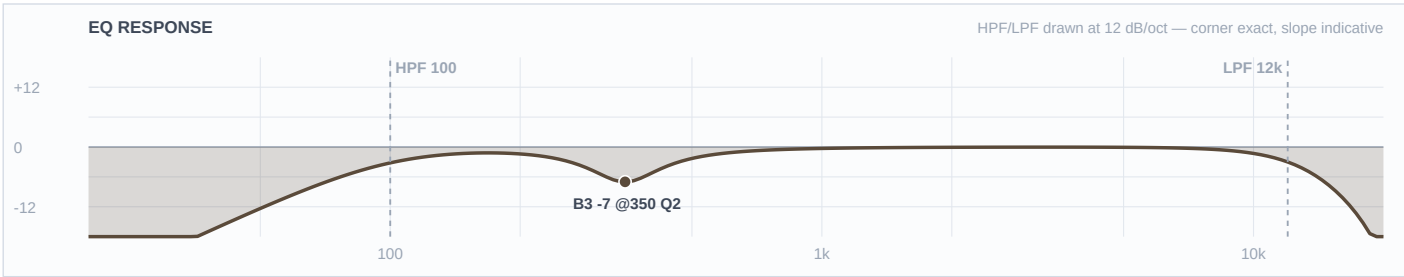
Mic Notes: SoundRef / Shure: flat 20Hz–20kHz with NO presence peak and no hyped top end — 'the brightness you get on playback is the brightness of the cymbal, not an artifact of the microphone.' That is precisely why the +3@10000 is legal here and why the KB's bias line for this mic is 'apply the template as-is.' Contrast rev 1, where the ND408's presence RISE made a 4500 cut load-bearing; this channel's 3500 cut is fixing the cymbal, not the mic. Cymbal group: HAT OWNS THE HIGH SIZZLE at 10k.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	250 Hz	LC	—	—	High-pass
4	10 kHz	Bell	+3 dB	1.5	
3	3.5 kHz	Bell	-6 dB	2	
2	400 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: In funk the hat is a lead rhythmic voice, not a garnish — sixteenths, open accents, the foot — and on this set it carries the groove's top layer for three hours. Because the SM81 flatters nothing, every move here is mine and every one is load-bearing. The 10k lift is the show's only high boost and it exists for a specific reason: 38 percent humidity means dry air strips high frequencies over the throw, so the sizzle that reads at the kit does not reach the back of the plaza on its own. The 3500 cut is the cymbal's own clank, moved off 4000 so it doesn't sit on top of the bottom-snare's cut at the same frequency. And -7@400 is snare bleed, which on a hat mic is the real enemy — a flat capsule reports all of it honestly.

Ch 6 | Rack 1

Mic: Audix D2



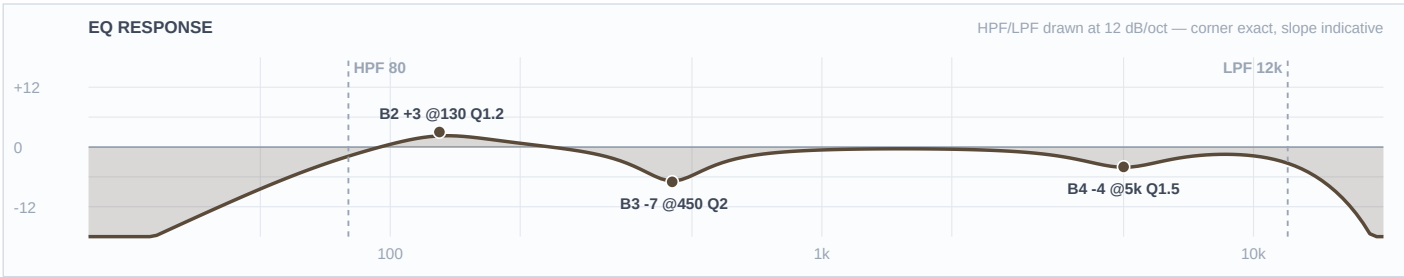
Mic Notes: Audix cutsheet: boost around 150Hz, a dip between 500Hz and 1kHz, and a subtle upper-mid presence lift, over 80Hz–18kHz. Both of rev 1's boosts came out because of that — +3@130 stacked on the baked 150 body, and +3@4000 stacked on the baked upper-mid lift. The box cut MOVED from 450 to 350 to stay outside the capsule's own 500Hz–1kHz dip. Tom section: RACK 1 OWNS BOX AT 350, and takes no boost at all.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	100 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	350 Hz	Bell	-7 dB	2	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: Toms are fill instruments in this band — they appear in turnarounds, not continuously — so the only thing that matters is that a fill lands clearly through a dance mix. This channel is deliberately the lightest on the kit, and that is a finding rather than an omission: Audix built the D2 for rack toms and reviewers are blunt that it delivers the modern rack-tom sound with no EQ at all, so the body and the attack are both already there. Even the ring cut a tom normally wants is unavailable, because tom ring on a small drum sits in the 600–900Hz window the capsule has already dipped. What is left is the shell box at 350, at full outdoor depth, placed below the dip where there is real response to remove — plus LPF 12k, which matters more than usual with three cymbal mics and a stereo overhead pair leaking into it.

Ch 7 | Rack 2

Mic: Audix D4



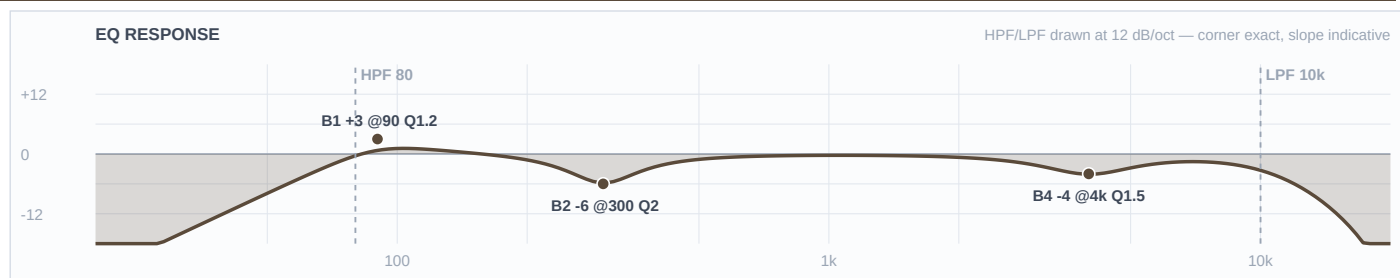
Mic Notes: RecordingHacks reading Audix's CURRENT chart: +6dB at 5kHz, a secondary peak above 10kHz, and a rolloff below 70Hz — so it does NOT lack attack and it does NOT reach into the 30s. The mic-library row was corrected to this on 2026-07-30 for exactly this reason; the old row would have put a click boost onto a +6dB baked peak. Weight is restored at 130, this drum's own note, where the chart documents no peak. Tom section: RACK 2 OWNS BOX AT 450 AND WEIGHT AT 130.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	80 Hz	LC	—	—	High-pass
4	5 kHz	Bell	-4 dB	1.5	
3	450 Hz	Bell	-7 dB	2	
2	130 Hz	Bell	+3 dB	1.2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: The middle drum, and the only tom that gets its pitch put back by the desk. The D4 is voiced HF-forward with a +6dB peak right where a tom's click lives, so the reflex boost becomes a -4 trim — and unlike the snare, this trim is NOT lightened for dry air, because on a ringy tom that region is ring rather than crack and the air does nothing to shorten ring. Underneath, the chart rolls off below 70Hz, so there is no sub reach to lean on outdoors and the honest move is to restore the drum's own fundamental at 130 rather than reach for weight that isn't there. 130 also keeps it off the kick and bass lanes and clear of the other two toms. The 450 box cut is offset from rack 1's 350 so the pair reads as two drums, not one smeared one.

Ch 8 | Floor

Mic: Audix D6



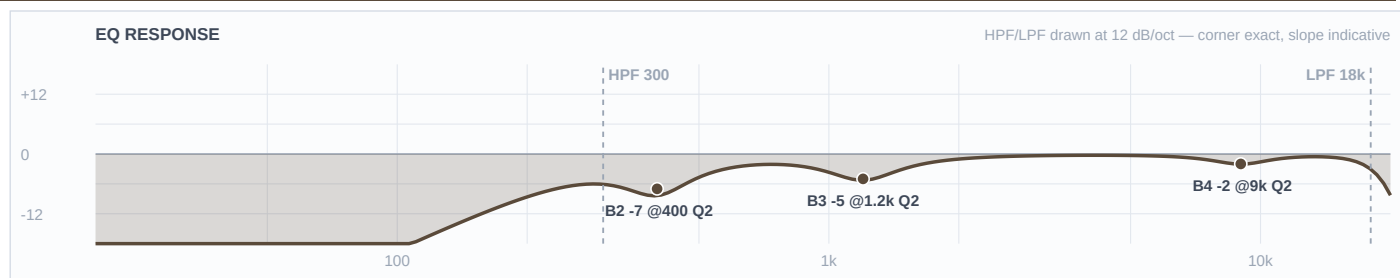
Mic Notes: Locker fork — MD 421-U offered, D6 kept (Brian's call). Tape Op's measured review: the low end peaks at 60Hz, the high end peaks at 4kHz AND 10kHz, and the midrange at 750Hz sits about 15dB below those peaks. B3 is therefore FLAT — cutting box where the capsule is already 15dB down digs an existing hole twice. HPF 80 is set ABOVE the 60Hz peak on purpose, to keep a kick capsule from making this drum read as a second kick; B1 +3@90 then restores the drum's own note, on unvoiced slope between the peak and the dip. LPF 10000 shades the capsule's second documented peak. Tom section: FLOOR OWNS RING AT 300 AND NOTE AT 90.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	80 Hz	LC	—	—	High-pass
4	4 kHz	Bell	-4 dB	1.5	
3	—	OFF	—	—	Flat
2	300 Hz	Bell	-6 dB	2	
1	90 Hz	Bell	+3 dB	1.2	
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: A kick capsule on a floor tom, kept deliberately, so this channel is built for that voicing rather than fighting it — Audix names floor tom as an intended application and reviewers report well-defined lows with a prominent stick sound, which is exactly the modern scooped dance-floor drum. The whole strategy is two moves in sequence: filter the capsule's 60Hz peak off the kick and bass lanes with an HPF at 80, then put the drum's actual note back at 90. Skip the first and the biggest punctuation drum in the fills turns into a second kick on a plaza with no room to absorb it. The 300 ring cut is available and needed because floor-tom ring sits well outside the 750Hz scoop — that band is what keeps this channel from being uselessly thin, given that its body was removed at the factory.

Ch 9 | Overheads

Mic: Shure Beta 27 (pair)

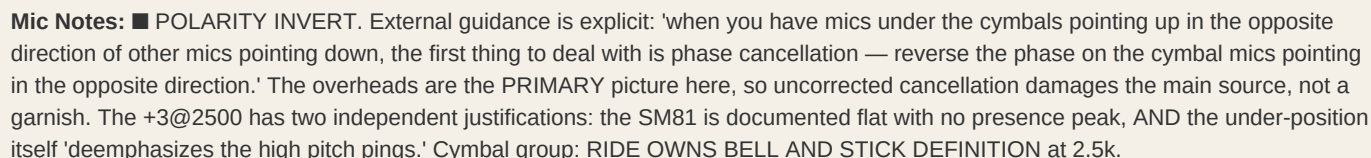


Mic Notes: RecordingHacks / Shure: nominally flat 60Hz–3kHz with two SMALL HF peaks of +2dB or less at 5500Hz and 9kHz, plus a –15dB pad and a 3-position LF filter; Shure names drum overheads as an optimal application. The +2dB figure is why the 9k trim is only –2 — rev 1's –4 removed more than the capsule added, which on a near-flat mic means cutting real cymbal air. Cymbal group: OVERHEADS ARE THE PRIMARY PICTURE and take no boost; the hat owns 10k, the ride owns 2.5k.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	300 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-2 dB	2	
3	1.2 kHz	Bell	-5 dB	2	
2	400 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	18 kHz	HC	—	—	Low-pass

Engineer Notes: The overheads are the actual cymbal picture on this show — the hat and the under-ride reinforce them rather than replace them — and they are also what makes a drummer sound like one instrument instead of nine channels. Two values moved off rev 1 purely on the weather: at 38 percent humidity the air is already attenuating the top over the throw, so the 9k trim came down to –2 to net a flat capsule back to roughly flat, and the LPF opened from 16k to 18k rather than closing off air the plaza is removing anyway. What is refused is the obvious response to dry air, an air boost up top: the hat already carries the show's only high lift, and two condensers lifted in the same region stack into hiss and cost feedback margin with four handhelds live. Below the cymbals this channel does nothing at all — HPF 300 and –7@400 — because everything down there arrives as bleed and every open condenser on this stage is a gain-before-feedback cost.

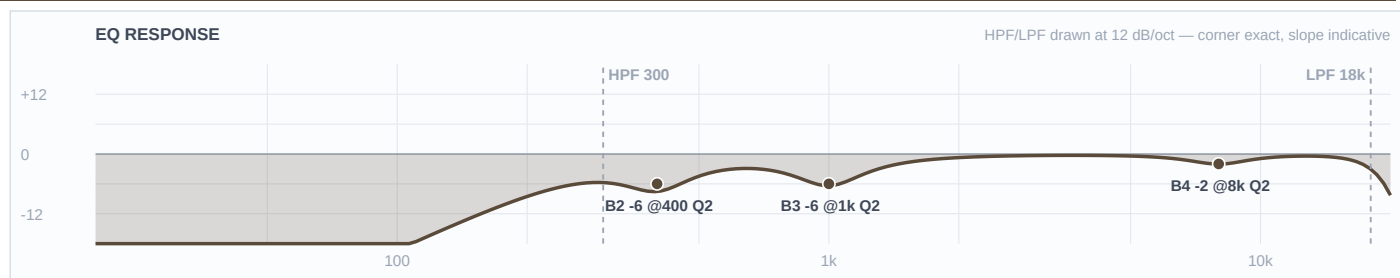
Mic: Shure SM81



Engineer Notes: A new channel this week, and an unusual one: from underneath you get stick definition and bell without the wash the top of a ride throws at the overheads, at the cost of a phase relationship that has to be fixed rather than mixed around. In this band the ride carries the jazz-inflected numbers and the half-time feels, and the bell is a hook in the gospel-adjacent material, so it earns a real lift — the strongest-justified boost on the show, because both the flat capsule and the position's own documented loss of ping point the same way, and dry air over the plaza throw makes a third argument. What it does not get is a high-sizzle boost; the hat owns that lane and a second lifted condenser would just add wash. Below, this mic is aimed up past the floor tom and snare, so 700 takes the underside gong character and 500 takes the bleed — offset from the overheads' 400 because from down here it hears different drums.

Ch 19 | Toys

Mic: Shure SM81



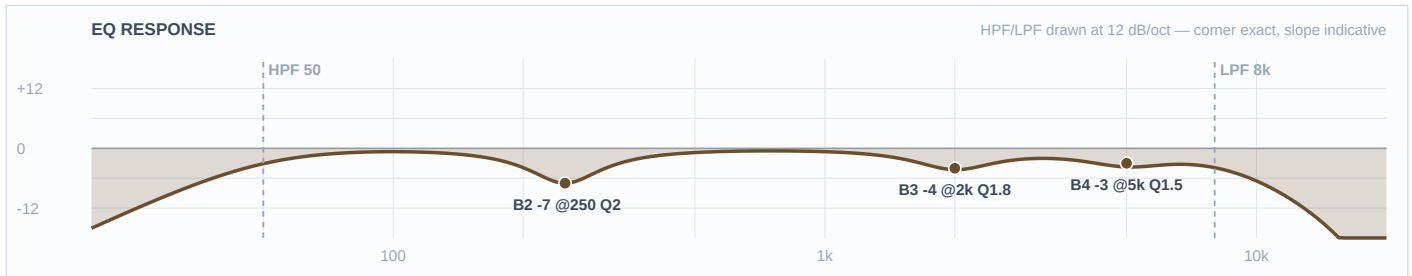
Mic Notes: Shure/MicrophoneGeeks, directly on point: 'percussion instruments with high-end detail — including tambourines, shakers and triangles — benefit from the smooth upper frequencies of the SM81.' Flat 20Hz–20kHz with no presence peak, so the 6–10kHz glare a shaker piles up is the INSTRUMENT's and legitimate to cut. Trim sized to –2 rather than rev 1's –3 by the dry-air read, and the LPF opened to 18k for the same reason. No high boost by sectional decision, not by capsule limit — the hat owns 10k and the ride owns 2.5k.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	300 Hz	LC	—	—	High-pass
4	8 kHz	Bell	-2 dB	2	
3	1 kHz	Bell	-6 dB	2	
2	400 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	18 kHz	HC	—	—	Low-pass

Engineer Notes: The only hand percussion left after the congas and bongos came off the list, which makes this the whole texture layer for a funk groove that wants to feel busy without adding another drum. Because the capsule flatters nothing, the sparkle is genuinely available for the taking — and it is deliberately not taken. Two other flat condensers on this stage already carry high lifts, and a third would stack into wash while costing feedback margin with four handhelds live, so the brightness comes from the LPF sitting at 18k rather than from a boost. What is cut is the tambourine and shaker glare piling up around 8k, lightened to –2 because dry air is already removing some of it over the throw, plus the woodblock honk at 1k and stage bleed at 400. HPF 300 because a toy tray has nothing musical below that and sits in the middle of a loud stage.

Ch 11 | Bass DI

Mic: XLR line feed



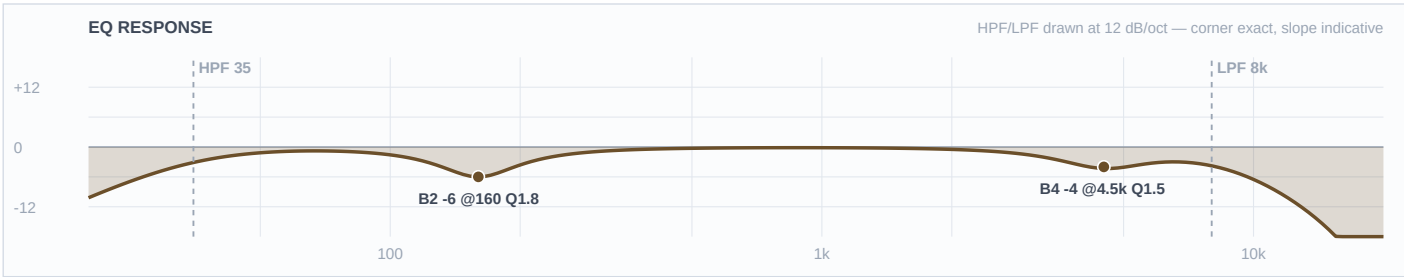
Mic Notes: POST-EQ FEED — DO NOT RE-VOICE. TalkBass consensus: a post DI is often excessively bright because 'a traditional bass speaker functions like a high pass filter and rolls off the highs — if the player cranks the highs to compensate, the sound coming out of the post DI will be obnoxiously bright', and the amp's baked curve has already cut its own mids. That is why rev 1's +3@2500 was withdrawn and replaced with trims. The 5000 trim is deliberately modest at -3 — 'often bright' is a tendency, not a measurement, so expect to want more and ride it at soundcheck. Two-mic pair with ch 12: CH 11 OWNS DEFINITION AND TOP. Polarity-check the pair in mono; flip ch 12 if the sum thins.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	50 Hz	LC	—	—	High-pass
4	5 kHz	Bell	-3 dB	1.5	
3	2 kHz	Bell	-4 dB	1.8	
2	250 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

Engineer Notes: The bass is the spine of every Motown walk and funk pocket in this set, and this leg is the one you follow the notes on. Because the feed is post-EQ, the job flips from creative to corrective: the player already decided the tone through their cab, and my adding presence on top of a signal they brightened to compensate for that cab's rolloff is how a bass turns to clank on a full-range PA. So all three bands are cuts. The 250 goes to full outdoor depth because bass mud on a plaza dance floor is what turns a line into a rumble, and it is also the biggest single favour I can do the four vocals. HPF at 50 keeps the low E's fundamental substantially intact while handing the deep weight to the cab mic, which is where the lane split puts it.

Ch 12 | Bass Mic

Mic: Shure PG52



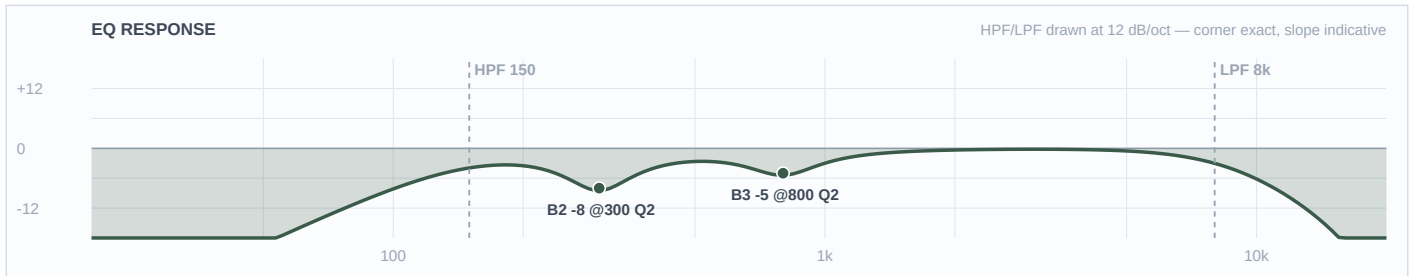
Mic Notes: RecordingHacks: 30Hz–13kHz cardioid, 1.8mV/Pa. Shure names close-miked bass amps as an intended application, and the successor PGA52's published description confirms the shape — 'a slight presence peak in the upper mids... with a scoop in the midsection and not a lot of upper top end.' All three KB bias rules are honoured literally: never boost the 60–100Hz hump (B1 FLAT — own the low end by letting it through), never cut into the 200–800Hz dip (B3 FLAT — the box cut sits at 160 instead), ease off presence at ~4500. Two-mic pair with ch 11: CH 12 OWNS THE LOW BODY. Polarity-check in mono; flip this channel if the sum thins.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	35 Hz	LC	—	—	High-pass
4	4.5 kHz	Bell	-4 dB	1.5	
3	—	OFF	—	—	Flat
2	160 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

Engineer Notes: This is the weight the KS21 arch moves air with, and it is also the one channel on the show allowed to reach — HPF 35, the lowest anywhere here, because the lane split makes it the designated bottom. Almost everything about the channel is a refusal. The low end is claimed by letting the capsule's own 60–100Hz hump through rather than boosting it, because boosting a voiced hump outdoors turns weight into boom with nothing to absorb it. The box cut cannot go where a box cut normally goes, since 200–800Hz is inside the mic's own broad dip, so it sits at 160 where there is real response to remove. And the 4500 trim takes the presence rise off, offset from ch 11's 5000 so the two legs of one instrument aren't shaped at the same frequency. It is a short list of moves because the PG52 on a bass cab is one of the few pairings in the locker where the curve is already the right shape.

Ch 13 | Guitar

Mic: XLR line feed



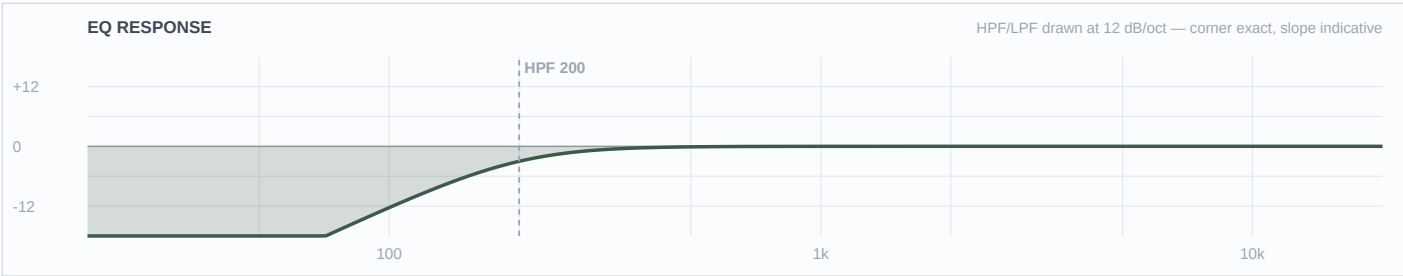
Mic Notes: EMULATED OUT = TREAT AS A MIC'D CAB. Marshall/Softube: the DSL's emulated output is a frequency-compensated line-level out running Softube's emulation of the Marshall 1960 4x12, on both the headphone and line outs. The capsule gate covers analog cab emulation exactly as it covers an IR, so B4 stays FLAT — the emulation already supplies the cab's presence shaping around 2–3kHz. Naming the cab is what moved the honk cut from rev 1's generic 900 to 800, the 1960's own midrange character. LPF 8000 is housekeeping only; the emulation already rolls off above ~5–6kHz.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	150 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	-5 dB	2	
2	300 Hz	Bell	-8 dB	2	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

Engineer Notes: Funk and Motown guitar is chunk — tight sixteenth comping, wah figures, the occasional solo — and it has to be articulate without eating the space four vocals need. The single most important thing on this channel is a boost that isn't here: a presence lift would have stacked straight onto the 1960 emulation's own 2–3kHz shaping, and the same withdrawal happened on rev 1 the moment cab sim was confirmed. So the articulation comes from clearing instead: -8@300 at full outdoor depth, because guitar mud stacks directly onto bass and keys mud in the same octave and a plaza is unforgiving about it, and -5@800 for the Marshall honk. Above 800 the channel is left alone deliberately — with no horn channels on this show, the keys and pads are playing the horn lines in the 1–2kHz window, and this guitar's job is to interlock with them rather than mask them.

Ch 15 | Click

Mic: XLR line feed



Mic Notes: Routing channel, not a tone channel. HPF 200 is housekeeping only. There is deliberately no EQ here: a click channel with EQ on it implies somebody once considered putting it in the PA.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	200 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	—	OFF	—	—	Flat
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: This channel exists so that it is accounted for and so nobody patches it into the PA by accident — that is the entire deliverable. The band's arrangements are locked to it and the Track rides underneath, so it has to be present and reliable at the drummer's end while being structurally incapable of reaching the square. It is muted from the mains, out of the record bus, and carries a single housekeeping HPF and nothing else.

Ch 16 | Track

Mic: XLR line feed



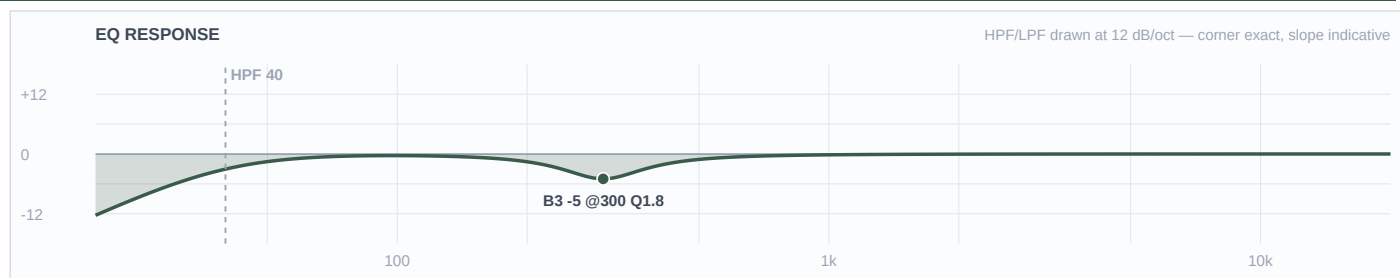
Mic Notes: Finished, mixed and mastered program material — not a source to shape. Every boost would be re-EQing someone else's master through a plaza PA, so there are none. The single 250Hz cut is room-making rather than tone: the master carries its own kick and bass energy sitting on ch 1/2 and ch 11/12 in that octave. HPF 35 lets the programmed sub content through — the KS21 arch reproduces it and it is part of the arrangement.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	35 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	250 Hz	Bell	-4 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: With no horn channels on the list, this bed and the keys and pads are the horn and string section, so the Track is load-bearing — if it goes away the arrangement has holes in it. That is exactly why it gets almost nothing done to it: it arrived finished, and the temptation on a plaza is to 'help' a master that is already balanced for full-range playback. The one cut at 250 is the collision that genuinely needs managing, because the master's own low end lands on top of a live kick pair and a live bass pair in the same octave, and something has to give way. HPF 35 rather than higher, because the programmed low end is a deliberate part of how this band sounds.

Ch 17 | Pad 1

Mic: BSS AR-133 (DI)



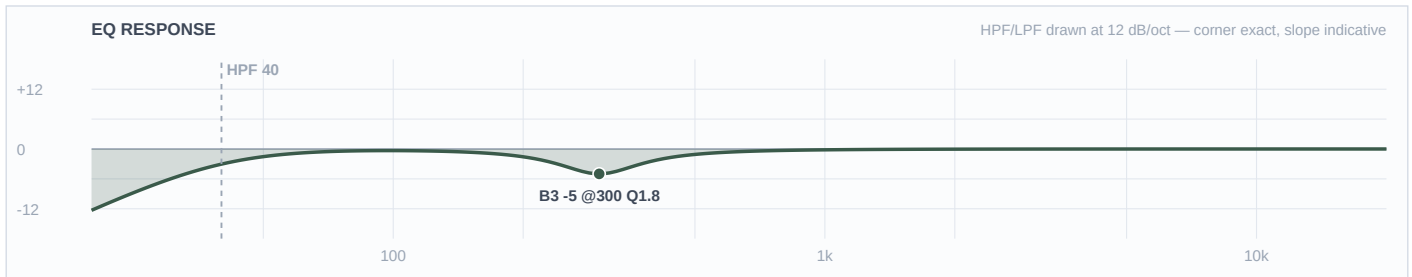
Mic Notes: Produced, pre-mixed material at line level — every sample was EQ'd and compressed before it reached the pad, so there is no capsule or transducer character to correct and no boost is available. The 300Hz cut is room-making against the guitar's -8 and the keys' -6 at the same octave, sized DOWN to -5 because deep-cutting finished material thins it. Identical values on ch 17 and 18 are correct: two halves of one instrument's output.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	40 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	300 Hz	Bell	-5 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: The pad shares horn-and-string duty with the keys and the Track, firing stabs, risers and vocal hits that are already produced — so this is a level-and-routing channel with one corrective move, not a tone channel. Its 1–2kHz window is left completely alone for the same reason the keys' is: cutting there would gut the programmed brass. What it does get is a share of the 300Hz clearance, because three sources landing in that octave on a plaza is precisely where a dance mix goes opaque, and this is one of the five instrument channels being cleared so four handhelds can sit forward without a single dB of vocal boost. HPF 40 keeps the samples' sub content, which the arch can reproduce.

Ch 18 | Pad 2

Mic: Whirlwind IMP (passive DI)



Mic Notes: See ch 17 — identical treatment by design. Produced material at line level, no transducer to correct, no boost available. Passive DI so no 48V. The pair is panned rather than summed; giving the two halves different curves would smear the pad's own image.

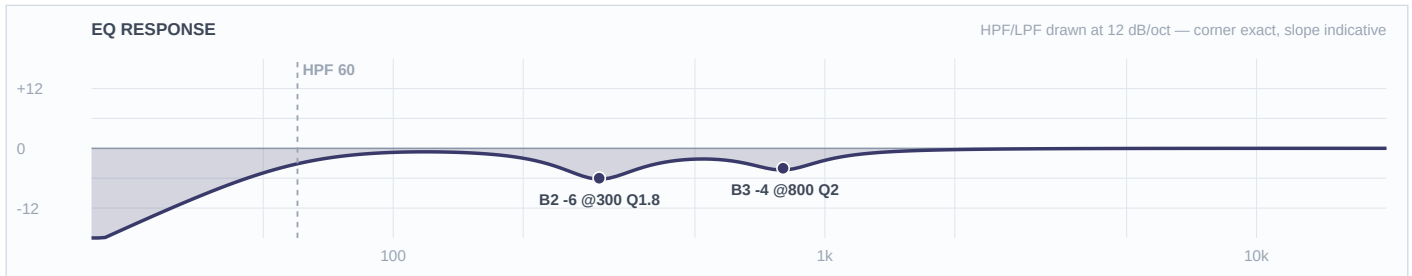
Band	Freq	Type	Gain	Q	Notes / Rationale
LC	40 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	300 Hz	Bell	-5 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: The other half of the sampling pad, treated identically on purpose. When a source is panned rather than summed, matching the two channels is what preserves the image the programmer built — this is one of only two places on the show where identical values are the correct answer rather than a slotting failure, the other being the keys. Everything in ch 17's reasoning applies here unchanged: produced samples get room-made for, not shaped, and the 300Hz clearance is part of buying vocal presence without touching a vocal fader.

■ KEYS ■

Ch 20 | Key L

Mic: XLR line feed



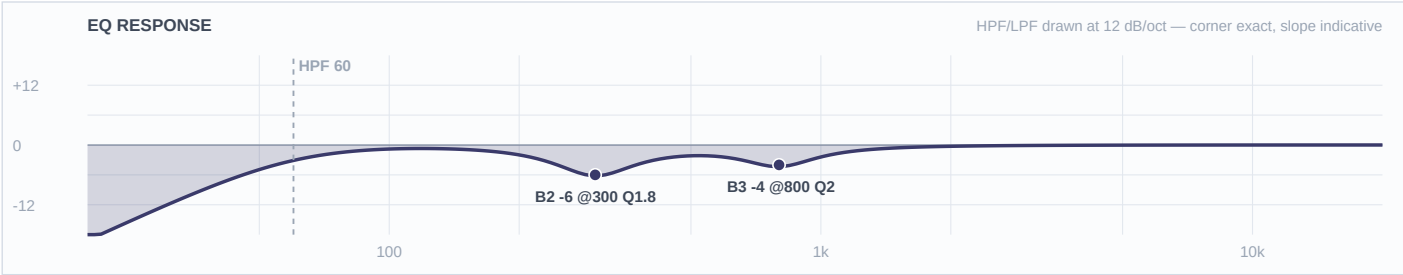
Mic Notes: THE 1–2kHz WINDOW IS LEFT UNTOUCHED ON PURPOSE — with no horn channels on this show, the keys carry horn and string lines and a cut there would gut them. That is the artist layer overriding a move the genre and venue layers would both otherwise call for. Patch voicing was chosen at the instrument, so no presence boost is available. The 800 honk cut is HELD to –4 rather than deepened, because the guitar already takes –5 at 800 and both channels cutting hard at the same frequency digs a hole where the comping interlocks. Identical values on 20 and 21 are correct — one instrument's stereo image.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	-4 dB	2	
2	300 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: The keys are doing two jobs at once here: the Rhodes, organ and piano comping an R&B and Motown set needs, and a share of the horn section that does not exist on this input list. That second job is what makes this the most restrained channel on the show. A stereo keyboard bed is the widest and most continuous thing in the mix and will mask a vocal before any drum does, so on almost any other gig the 1–2kHz window is the first place I would go — and here it is off limits, because that is where the brass lines live. So the work happens around it: –6@300 at real outdoor depth for the accumulation this instrument shares with the guitar, bass and pads, a modest –4@800 that deliberately stops short of the guitar's cut at the same frequency, and HPF 60 so a piano patch keeps its low register without trespassing on the kick and bass.

Ch 21 | Key R

Mic: XLR line feed



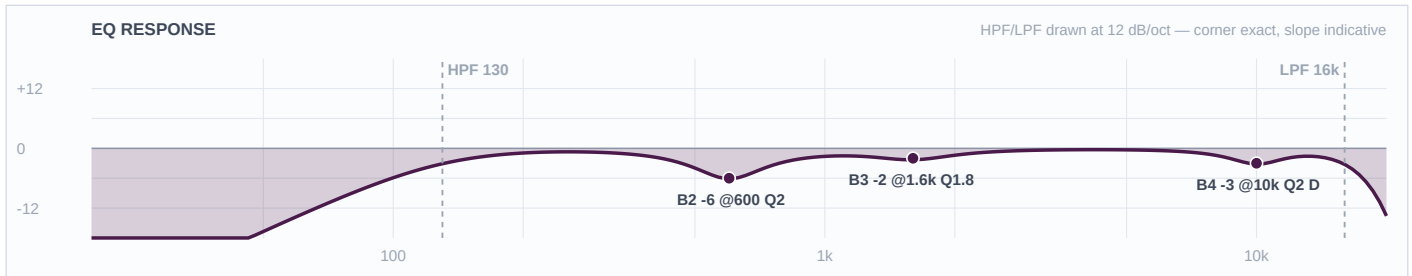
Mic Notes: See ch 20 — identical by design, because this is the right half of one stereo image and differing curves would smear it. The 1–2kHz window stays untouched on both sides for the horn lines. No presence boost: the patch voicing is upstream.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	-4 dB	2	
2	300 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: The right half of the keyboard rig, matched to ch 20 on purpose. Matching is the correct answer for a genuine stereo source in a way it never is for members of a section — the three toms and the four vocals get separated in the numbers, but splitting a piano's image between two different curves just makes the instrument sound broken. Everything in ch 20's reasoning holds here: the horn-line window is protected, the 300 clearance is part of buying vocal headroom across five instrument channels, and 800 stays modest so it does not stack with the guitar's cut at the same spot.

Ch 33 | Aretha

Mic: Shure Beta 58A (Wireless 1)



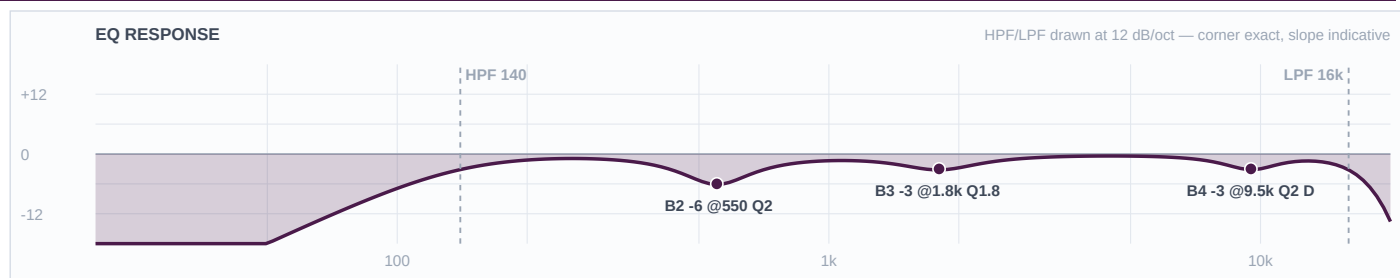
Mic Notes: Shure/MusicOnStage: TWO presence peaks, at 4kHz and 10kHz, with a pronounced 4–9kHz region far more aggressive than an SM58's, and frequencies attenuated below 500Hz to counter proximity effect. Cuts only — the rule, and also nothing to add. The de-esser is DYNAMIC because the 10kHz peak only misbehaves on sibilants, and a static value set at 7pm in 38 percent RH would be wrong by 10pm at 60. B2 uses a tight Q2.0 deliberately: the capsule already attenuates below 500, so a broad cut would double-dip that rolloff. Vocal slot: HPF 130 · box 600 · mid 1600 · de-ess 10000.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	130 Hz	LC	—	—	High-pass
4	10 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.6 kHz	Bell	-2 dB	1.8	
2	600 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The lead voice and the MC, so this channel is live and talking between songs as well as singing, on the most exposed position of the four. Everything is subtractive, because with four handhels on an open plaza gain-before-feedback is the whole game and because the Beta 58A's twin presence peaks mean there is nothing worth adding anyway. Her presence gets bought elsewhere: five instrument channels are cut in her masking region so she sits forward without a vocal fader moving. Her slot is the highest de-esser of the four at 10k and a box cut at 600, both placed to stay off the other three, and the 1600 cut is only -2 because as the lead she needs the most intelligibility left intact of anyone. The template's 184 HPF and its 5k feedback notch are both gone, which means the feedback protection has to be re-earned by hand at soundcheck.

Ch 34 | Heather

Mic: Shure Beta 58A (Wireless 2)



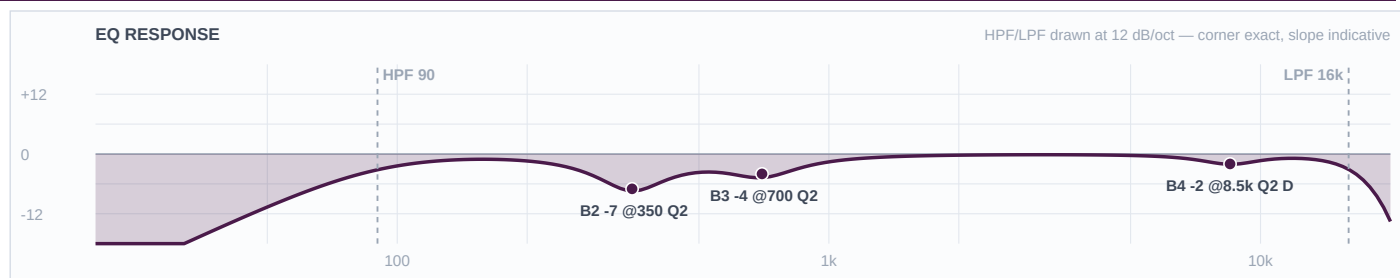
Mic Notes: Same capsule as ch 33/35/36 — 4kHz and 10kHz presence peaks, sub-500Hz attenuation. Cuts only, dynamic de-esser, tight-Q box cut. Vocal slot: HPF 140 · box 550 · mid 1800 · de-ess 9500. No value is shared with any other vocal channel.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	140 Hz	LC	—	—	High-pass
4	9.5 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.8 kHz	Bell	-3 dB	1.8	
2	550 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The second female voice, and the closest of the four to Aretha, which is exactly why the slotting matters most here — two women trading and stacking harmonies on the same capsule model will mask each other before anything in the band does. So she sits above Aretha on the HPF at 140, below her on the box at 550, above her on the nasal cut at 1800 and below her on the de-esser at 9500: adjacent in every band, identical in none. Her 1800 cut is a full -3 rather than Aretha's -2 because she is not carrying the between-song talking and can afford slightly more nasal control. Like all four, no boosts anywhere, dynamic de-esser rather than static so it tracks the humidity climbing 22 points across the set.

Ch 35 | Vince

Mic: Shure Beta 58A (Wireless 3)



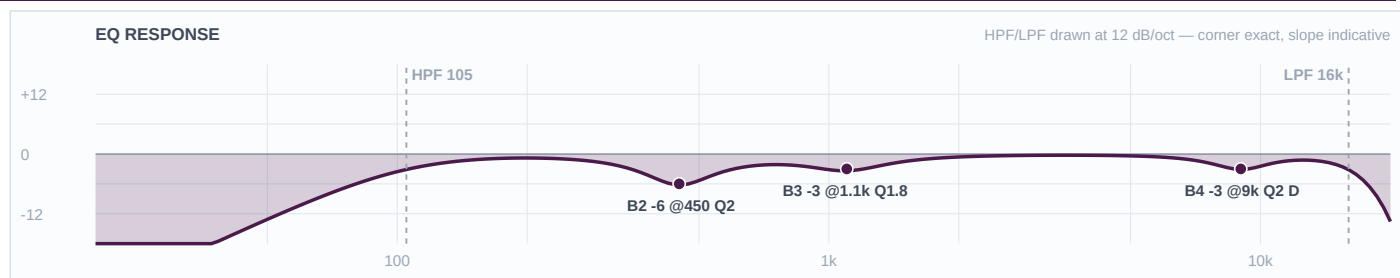
Mic Notes: Same Beta 58A capsule facts. THE UPPER-MID CUT SITS AT 700Hz — BELOW his presence region — on purpose: for a bass voice, cutting inside the 1.2–1.8kHz nasal window removes the intelligibility he depends on, which is why the other three are cut in there and he is not. HPF 90 because his fundamentals reach ~82Hz; the template's 184 would have removed the bottom of his voice. Lightest de-esser of the four at -2, and the lowest frequency at 8500. Vocal slot: HPF 90 · box 350 · mid 700 · de-ess 8500.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	90 Hz	LC	—	—	High-pass
4	8.5 kHz	Bell	-2 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	700 Hz	Bell	-4 dB	2	
2	350 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The bass voice, and the channel where the template's baseline was most dangerous — an HPF at 184 on a singer whose fundamentals reach into the low 80s removes the part of him that makes him worth featuring. So he gets the lowest HPF of the four at 90 and the deepest box cut at 350, which is the one place a low male voice genuinely piles up on a plaza. The move that matters most is where his upper-mid cut is NOT: everyone else is cut inside 1.2–1.8kHz because their presence regions sit higher, and cutting him there would take the intelligibility a bass voice has least of to spare, so his sits at 700 instead. His de-esser is the lightest and lowest of the four, because a bass-baritone's sibilance arrives lower and quieter than a soprano's and the Beta 58A's 10k peak has less to grab.

Ch 36 | Brandon

Mic: Shure Beta 58A (Wireless 4)



Mic Notes: ■ RANGE UNKNOWN — built as a deliberately offset middle-of-the-road male voice, placed BETWEEN Vince and the two women in every band so that whichever way his voice turns out he is not colliding with an already-slotted channel. Same Beta 58A capsule facts as 33/34/35: cuts only, dynamic de-esser, tight-Q box. Both alternate sets are in the notes above and neither displaces another singer's slot. Vocal slot: HPF 105 · box 450 · mid 1100 · de-ess 9000.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	105 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.1 kHz	Bell	-3 dB	1.8	
2	450 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: A new person on a fader that was somebody else last week, and the honest position is that I have not heard him — so rather than dress a guess up as a value, this channel is built in the gap. HPF 105 sits between Vince's 90 and Aretha's 130, box 450 between Vince's 350 and Heather's 550, nasal cut 1100 just below the women's 1600 and 1800 and well above Vince's 700. That placement is the point: it is defensible for a baritone, a tenor or anything between, and it collides with nobody. Both alternate sets are printed on this page so the change is a 20-second job at the desk once he opens his mouth, and either one still leaves all four singers on unique values. Cuts only and a dynamic de-esser, same as the other three.